

VIII.

Criticism and Counterproposals to the Copenhagen Interpretation of Quantum Theory

THE Copenhagen interpretation of quantum theory has led the physicists far away from the simple materialistic views that prevailed in the natural science of the nineteenth century. Since these views had not only been intrinsically connected with natural science of that period but had also found a systematic analysis in some philosophic systems and had penetrated deeply into the mind even of the common men on the street, it can be well understood that many attempts have been made to criticize the Copenhagen interpretation and to replace it by one more in line with the concepts of classical physics or materialistic philosophy.

These attempts can be divided into three different groups. The first group does not want to change the Copenhagen interpretation so far as predictions of experimental results are concerned; but it tries to change the language of this interpretation in order to get a closer resemblance to classical physics. In other words, it tries to change the philosophy without changing the physics. Some papers of this first group restrict their agreement

with the experimental predictions of the Copenhagen interpretation to all those experiments that have hitherto been carried out or that belong to normal electronic physics.

The second group realizes that the Copenhagen interpretation is the only adequate one, if the experimental results agree everywhere with the predictions of this interpretation. Therefore, the papers of this group try to change quantum theory to some extent in certain critical points.

The third group, finally, expresses rather its general dissatisfaction with the results of the Copenhagen interpretation and especially with its philosophical conclusions, without making definite counterproposals. Papers by Einstein, von Laue and Schrödinger belong to this third group which has historically been the first of the three groups.

However, all the opponents of the Copenhagen interpretation do agree on one point. It would, in their view, be desirable to return to the reality concept of classical physics or, to use a more general philosophic term, to the ontology of materialism. They would prefer to come back to the idea of an objective real world whose smallest parts exist objectively in the same sense as stones or trees exist, independently of whether or not we observe them.

This, however, is impossible or at least not entirely possible because of the nature of the atomic phenomena, as has been discussed in some of the earlier chapters. It cannot be our task to formulate wishes as to how the atomic phenomena should be; our task can only be to understand them.

When one analyzes the papers of the first group, it is important to realize from the beginning that their interpretations cannot be refuted by experiment, since they only repeat the Copenhagen interpretation in a different language. From a strictly positivistic standpoint one may even say that we are here

concerned not with counterproposals to the Copenhagen interpretation but with its exact repetition in a different language. Therefore, one can only dispute the suitability of this language. One group of counterproposals works with the idea of "hidden parameters." Since the quantum-theoretical laws determine in general the results of an experiment only statistically, one would from the classical standpoint be inclined to think that there exist some hidden parameters which escape observation in any ordinary experiment but which determine the outcome of the experiment in the normal causal way. Therefore, some papers try to construct such parameters within the framework of quantum mechanics.

Along this line, for instance, Bohm has made a counterproposal to the Copenhagen interpretation, which has recently been taken up to some extent also by de Broglie. Bohm's interpretation has been worked out in detail. It may therefore serve here as a basis for the discussions. Bohm considers the particles as "objectively real" structures, like the point masses in Newtonian mechanics. The waves in configuration space are in his interpretation "objectively real" too, like electric fields. Configuration space is a space of many dimensions referring to the different co-ordinates of all the particles belonging to the system. Here we meet a first difficulty: what does it mean to call waves in configuration space "real"? This space is a very abstract space. The word "real" goes back to the Latin word "res," which means "thing"; but things are in the ordinary three-dimensional space, not in an abstract configuration space. One may call the waves in configuration space "objective" when one wants to say that these waves do not depend on any observer; but one can scarcely call them "real" unless one is willing to change the meaning of the word. Bohm goes on defining the

lines perpendicular to the surfaces of constant wave-phase as the possible orbits of the particles. Which of these lines is the “real” orbit depends, according to him, on the history of the system and the measuring apparatus and cannot be decided without knowing more about the system and the measuring equipment than actually can be known. This history contains in fact the hidden parameters, the “actual orbit” before the experiment started.

One consequence of this interpretation is, as Pauli has emphasized, that the electrons in the ground states of many atoms should be at rest, not performing any orbital motion around the atomic nucleus. This looks like a contradiction of the experiments, since measurements of the velocity of the electrons in the ground state (for instance, by means of the Compton effect) reveal always a velocity distribution in the ground state, which is—in conformity with the rules of quantum mechanics—given by the square of the wave function in momentum or velocity space. But here Bohm can argue that the measurement can no longer be evaluated by the ordinary laws. He agrees that the normal evaluation of the measurement would indeed lead to a velocity distribution; but when the quantum theory for the measuring equipment is taken into account—especially some strange quantum potentials introduced ad hoc by Bohm—then the statement is admissible that the electrons “really” always are at rest. In measurements of the position of the particle, Bohm takes the ordinary interpretation of the experiments as correct; in measurements of the velocity he rejects it. At this price Bohm considers himself able to assert: “We do not need to abandon the precise, rational and objective description of individual systems in the realm of quantum theory.” This objective description, however, reveals itself as a kind of “ideological super-

structure," which has little to do with immediate physical reality; for the hidden parameters of Bohm's interpretation are of such a kind that they can never occur in the description of real processes, if quantum theory remains unchanged.

In order to escape this difficulty, Bohm does in fact express the hope that in future experiments in the range of the elementary particles the hidden parameters may yet play a physical part, and that quantum theory may thus be proved false. When such strange hopes were expressed, Bohr used to say that they were similar in structure to the sentence: "We may hope that it will later turn out that sometimes $2 \times 2 = 5$, for this would be of great advantage for our finances." Actually the fulfillment of Bohm's hopes would cut the ground from beneath not only quantum theory but also Bohm's interpretation. Of course it must at the same time be emphasized that the analogy just mentioned, although complete, does not represent a logically compelling argument against a possible future alteration of quantum theory in the manner suggested by Bohm. For it would not be fundamentally unimaginable that, for example, a future extension of mathematical logic might give a certain meaning to the statement that in exceptional cases $2 \times 2 = 5$, and it might even be possible that this extended mathematics would be of use in calculations in the field of economics. We are nevertheless actually convinced, even without cogent logical grounds, that such changes in mathematics would be of no help to us financially. Therefore, it is very difficult to understand how the mathematical proposals which the work of Bohm indicates as a possible realization of his hopes could be used for the description of physical phenomena.

If we disregard this possible alteration of quantum theory, then Bohm's language, as we have already pointed out, says

nothing about physics that is different from what the Copenhagen interpretation says. There then remains only the question of the suitability of this language. Besides the objection already made that in speaking of particle orbits we are concerned with a superfluous "ideological superstructure," it must be particularly mentioned here that Bohm's language destroys the symmetry between position and velocity which is implicit in quantum theory; for the measurements of position Bohm accepts the usual interpretation, for the measurements of velocity or momentum he rejects it. Since the symmetry properties always constitute the most essential features of a theory, it is difficult to see what would be gained by omitting them in the corresponding language. Therefore, one cannot consider Bohm's counterproposal to the Copenhagen interpretation as an improvement.

A similar objection can be raised in a somewhat different form against the statistical interpretations put forward by Bopp and (on a slightly different line) by Fenyés. Bopp considers the creation or the annihilation of a particle as the fundamental process of quantum theory, the particle is "real" in the classical sense of the word, in the sense of materialistic ontology, and the laws of quantum theory are considered as a special case of correlation statistics for such events of creation and annihilation. This interpretation, which contains many interesting comments on the mathematical laws of quantum theory, can be carried out in such a manner that it leads, as regards the physical consequences, to exactly the same conclusions as the Copenhagen interpretation. So far it is, in the positivistic sense, isomorphic with it, as is Bohm's. But in its language it destroys the symmetry between particles and waves that otherwise is a characteristic feature of the mathematical scheme of quantum theory. As early as 1928 it was shown by Jordan, Klein and Wigner that the

mathematical scheme can be interpreted not only as a quantization of particle motion but also as a quantization of three-dimensional matter waves; therefore, there is no reason to consider these matter waves as less real than the particles. The symmetry between waves and particles could be ensured in Bopp's interpretation only if the corresponding correlation statistics were developed for matter waves in space and time as well, and if the question was left open whether particles or waves are to be considered as the "actual" reality.

The assumption that particles are real in the sense of the materialistic ontology will always lead to the temptation to consider deviations from the uncertainty principle as "basically" possible. Fenyés, for instance, says that "the existence of the uncertainty principle [which he connects with certain statistical relations] by no means renders impossible the simultaneous measurement, with arbitrary accuracy, of position and velocity." Fenyés does not, however, state how such measurements should be carried out in practice, and therefore his considerations seem to remain abstract mathematics.

Weizel, whose counterproposals to the Copenhagen interpretation are akin to those of Bohm and Fenyés, relates the "hidden parameters" to a new kind of particle introduced ad hoc, the "zeron," which is not otherwise observable. However, such a concept runs into the danger that the interaction between the real particles and the zeron dissipates the energy among the many degrees of freedom of the zeron field, so that the whole of thermodynamics becomes a chaos. Weizel has not explained how he hopes to avoid this danger.

The standpoint of the entire group of publications mentioned so far can perhaps best be defined by recalling a similar discussion of the theory of special relativity. Anyone who was dis-

satisfied with Einstein's negation of the ether, of absolute space and of absolute time could then argue as follows: The non-existence of absolute space and absolute time is by no means proved by the theory of special relativity. It has been shown only that true space and true time do not occur directly in any ordinary experiment; but if this aspect of the laws of nature has been correctly taken into account, and thus the correct "apparent" times have been introduced for moving co-ordinate systems, there would be no arguments against the assumption of an absolute space. It would even be plausible to assume that the center of gravity of our galaxy is (at least approximately) at rest in absolute space. The critic of the special theory of relativity might add that we may hope that future measurements will allow the unambiguous definition of absolute space (that is, of the "hidden parameter" of the theory of relativity) and that the theory of relativity will thus be refuted.

It is seen at once that this argument cannot be refuted by experiment, since it as yet makes no assertions which differ from those of the theory of special relativity. But such an interpretation would destroy in the language used the decisive symmetry property of the theory, namely, the Lorentz invariance, and it must therefore be considered inappropriate.

The analogy to quantum theory is obvious. The laws of quantum theory are such that the "hidden parameters," invented ad hoc, can never be observed. The decisive symmetry properties are thus destroyed if we introduce the hidden parameters as a fictitious entity into the interpretation of the theory.

The work of Blochinzev and Alexandrov is quite different in its statement of the problem from those discussed before. These authors expressly and from the beginning restrict their objections against the Copenhagen interpretation to the philosophical side

of the problem. The physics of this interpretation is accepted unreservedly.

The external form of the polemic, however, is so much the sharper: "Among the different idealistic trends in contemporary physics the so-called Copenhagen school is the most reactionary. The present article is devoted to the unmasking of the idealistic and agnostic speculations of this school on the basic problems of quantum physics," writes Blochinzev in his introduction. The acerbity of the polemic shows that here we have to do not with science alone but with a confession of faith, with adherence to a certain creed. The aim is expressed at the end with a quotation from the work of Lenin: "However marvellous, from the point of view of the common human intellect, the transformation of the unweighable ether into weighable material, however strange the electrons lack of any but electromagnetic mass, however unusual the restriction of the mechanical laws of motion to but one realm of natural phenomena and their subordination to the deeper laws of electromagnetic phenomena, and so on—all this is but another *confirmation* of dialectic materialism." This latter statement seems to make Blochinzev's discussion about the relation of quantum theory to the philosophy of dialectic materialism less interesting in so far as it seems to degrade it to a staged trial in which the verdict is known before the trial has begun. Still it is important to get complete clarity about the arguments brought forward by Blochinzev and Alexandrov.

Here, where the task is to rescue materialistic ontology, the attack is chiefly made against the introduction of the observer into the interpretation of quantum theory. Alexandrov writes: "We must therefore understand by 'result of measurement' in quantum theory only the objective effect of the interaction of the electron with a suitable object. Mention of the observer must

be avoided, and we must treat objective conditions and objective effects. A physical quantity is an objective characteristic of the phenomenon, but not the result of an observation." According to Alexandrov, the wave function in configuration space characterizes the objective state of the electron.

In his presentation Alexandrov overlooks the fact that the formalism of quantum theory does not allow the same degree of objectivation as that of classical physics. For instance, if the interaction of a system with the measuring apparatus is treated as a whole according to quantum mechanics and if both are regarded as cut off from the rest of the world, then the formalism of quantum theory does not as a rule lead to a definite result; it will not lead, e.g., to the blackening of the photographic plate at a given point. If one tries to rescue Alexandrov's "objective effect" by saying that "in reality" the plate is blackened at a given point after the interaction, the rejoinder is that the quantum mechanical treatment of the closed system consisting of electron, measuring apparatus and plate is no longer being applied. It is the "factual" character of an event describable in terms of the concepts of daily life which is not without further comment contained in the mathematical formalism of quantum theory, and which appears in the Copenhagen interpretation by the introduction of the observer. Of course the introduction of the observer must not be misunderstood to imply that some kind of subjective features are to be brought into the description of nature. The observer has, rather, only the function of registering decisions, i.e., processes in space and time, and it does not matter whether the observer is an apparatus or a human being; but the registration, i.e., the transition from the "possible" to the "actual," is absolutely necessary here and cannot be omitted from the interpretation of quantum theory. At this point quan-

tum theory is intrinsically connected with thermodynamics in so far as every act of observation is by its very nature an irreversible process; it is only through such irreversible processes that the formalism of quantum theory can be consistently connected with actual events in space and time. Again the irreversibility is—when projected into the mathematical representation of the phenomena—a consequence of the observer's incomplete knowledge of the system and in so far not completely "objective."

Blochinzhev formulates matter slightly differently from Alexandrov: "In quantum mechanics we describe not a state of the particle in itself but the fact that the particle belongs to this or that statistical assembly. This belonging is completely objective and does not depend on statements made by the observer." Such formulations, however, take us very far—probably too far—away from materialistic ontology. To make this point clear it is useful to recall how this belonging to a statistical assembly is used in the interpretation of classical thermodynamics. If an observer has determined the temperature of a system and wants to draw from his results conclusions about the molecular motions in the system he is able to say that the system is just one sample out of a canonical ensemble and thus he may consider it as possibly having different energies. "In reality"—so we would conclude in classical physics—the system has only one definite energy at a given time, and none of the others is realized. The observer has been deceived if he considered a different energy at that moment as possible. The canonical ensemble contains statements not only about the system itself but also about the observer's incomplete knowledge of the system. If Blochinzev in quantum theory tries to call a system's belonging to an assembly "completely objective," he uses the word "objective" in a different

sense from that in classical physics. For in classical physics this belonging means, as has been said, statements not only about the system but also about the observer's degree of knowledge of the system. One exception must be made to this assertion in quantum theory. If in quantum theory the assembly is characterized by only one wave function in configuration space (and not, as usual, by a statistical matrix), we meet a special situation (the so-called "pure case") in which the description can be called objective in some sense and in which the element of incomplete knowledge does not occur immediately. But since every measurement would (on account of its irreversible features) reintroduce the element of incomplete knowledge, the situation would not be fundamentally different.

Above all, we see from these formulations how difficult it is when we try to push new ideas into an old system of concepts belonging to an earlier philosophy—or, to use an old metaphor, when we attempt to put new wine into old bottles. Such attempts are always distressing, for they mislead us into continually occupying ourselves with the inevitable cracks in the old bottles instead of rejoicing over the new wine. We cannot possibly expect those thinkers who a century ago introduced dialectic materialism to have foreseen the development of quantum theory. Their concepts of matter and reality could not possibly be adapted to the results of the refined experimental technique of our days.

Perhaps one should add at this point some general remarks about the attitude of the scientist to a special creed; it may be a religious or a political creed. The fundamental difference between the religious and the political creed—that the latter refers to the immediate material reality of the world around us, while the former has as its object another reality beyond the material

world—is not important for this special question; it is the problem of creed itself that is to be discussed. From what has been said one would be inclined to demand that the scientist should never rely on special doctrines, never confine his method of thinking to a special philosophy. He should always be prepared to have the foundations of his knowledge changed by new experience. But this demand would again be an oversimplification of our situation in life for two reasons. The first is that the structure of our thinking is determined in our youth by ideas which we meet at that time or by getting into contact with strong personalities from whom we learn. This structure will form an integrating part of all our later work and it may well make it difficult for us to adapt ourselves to entirely different ideas later on. The second reason is that we belong to a community or a society. This community is kept together by common ideas, by a common scale of ethical values, or by a common language in which one speaks about the general problems of life. The common ideas may be supported by the authority of a church, a party or the state and, even if this is not the case, it may be difficult to go away from the common ideas without getting into conflict with the community. Yet the results of scientific thinking may contradict some of the common ideas. Certainly it would be unwise to demand that the scientist should generally not be a loyal member of his community, that he should be deprived of the happiness that may come from belonging to a community, and it would be equally unwise to desire that the common ideas of society which from the scientific point of view are always simplifications should change instantaneously with the progress of scientific knowledge, that they should be as variable as scientific theories must necessarily be. Therefore, at this point we come back even in our time to the old problem of

the "twofold truth" that has filled the history of Christian religion throughout the later Middle Ages. There is the very disputable doctrine that "positive religion—whatever form it may take—is an indispensable need for the mass of the people, while the man of science seeks the real truth back of religion and seeks it only there." "Science is esoteric," so it is said, "it is only for the few." If in our time political doctrines and social activities take the part of positive religion in some countries, the problem is still essentially the same. The scientist's first claim will always be intellectual honesty, while the community will frequently ask of the scientist that—in view of the variability of science—he at least wait a few decades before expressing in public his dissenting opinions. There is probably no simple solution to this problem, if tolerance alone is not sufficient; but some consolation may come from the fact that it is certainly an old problem belonging to human life.

Coming back now to the counterproposals to the Copenhagen interpretation of quantum theory we have to discuss the second group of proposals, which try to change quantum theory in order to arrive at a different philosophical interpretation. The most careful attempt in this direction has been made by Janossy, who has realized that the rigorous validity of quantum mechanics compels us to depart from the reality concept of classical physics. He therefore seeks to alter quantum mechanics in such a way that, although many of the results remain true, its structure approaches that of classical physics. His point of attack is what is called "the reduction of wave packets," i.e., the fact that the wave function or, more generally, the probability function changes discontinuously when the observer takes cognizance of a result of measurement. Janossy notices that this reduction cannot be deduced from the differential equations of the mathe-

mathematical formalism and he believes that he can conclude from this that there is an inconsistency in the usual interpretation. It is well known that the "reduction of wave packets" always appears in the Copenhagen interpretation when the transition is completed from the possible to the actual. The probability function, which covered a wide range of possibilities, is suddenly reduced to a much narrower range by the fact that the experiment has led to a definite result, that actually a certain event has happened. In the formalism this reduction requires that the so-called interference of probabilities, which is the most characteristic phenomenon of quantum theory, is destroyed by the partly undefinable and irreversible interactions of the system with the measuring apparatus and the rest of the world. Janossy now tries to alter quantum mechanics by the introduction of so-called damping terms into the equations, in such a way that the interference terms disappear of themselves after a finite time. Even if this corresponds to reality—and there is no reason to suppose this from the experiments that have been performed—there would still remain a number of alarming consequences of such an interpretation, as Janossy himself points out (e.g., waves which are propagated faster than the velocity of light, interchange of the time sequence of cause and effect, etc.). Therefore, we should hardly be ready to sacrifice the simplicity of quantum theory for this kind of view until we are compelled by experiments to do so.

Among the remaining opponents of what is sometimes called the "orthodox" interpretation of quantum theory, Schrödinger has taken an exceptional position inasmuch as he would ascribe the "objective reality" not to the particles but to the waves and is not prepared to interpret the waves as "probability waves only." In his paper entitled "Are There Quantum Jumps?" he

attempts to deny the existence of quantum jumps altogether (one may question the suitability of the term "quantum jump" at this place and could replace it by the less provocative term "discontinuity"). Now, Schrödinger's work first of all contains some misunderstanding of the usual interpretation. He overlooks the fact that only the waves in configuration space (or the "transformation matrices") are probability waves in the usual interpretation, while the three-dimensional matter waves or radiation waves are not. The latter have just as much and just as little "reality" as the particles; they have no direct connection with probability waves but have a continuous density of energy and momentum, like an electromagnetic field in Maxwell's theory. Schrödinger therefore rightly emphasizes that at this point the processes can be conceived of as being more continuous than they usually are. But this interpretation cannot remove the element of discontinuity that is found everywhere in atomic physics; any scintillation screen or Geiger counter demonstrates this element at once. In the usual interpretation of quantum theory it is contained in the transition from the possible to the actual. Schrödinger himself makes no counterproposal as to how he intends to introduce the element of discontinuity, everywhere observable, in a different manner from the usual interpretation.

Finally, the criticism which Einstein, Laue and others have expressed in several papers concentrates on the question whether the Copenhagen interpretation permits a unique, objective description of the physical facts. Their essential arguments may be stated in the following terms: The mathematical scheme of quantum theory seems to be a perfectly adequate description of the statistics of atomic phenomena. But even if its statements about the probability of atomic events are completely correct, this interpretation does not describe what actually happens inde-

pendently of or between the observations. But something must happen, this we cannot doubt; this something need not be described in terms of electrons or waves or light quanta, but unless it is described somehow the task of physics is not completed. It cannot be admitted that it refers to the act of observation only. The physicist must postulate in his science that he is studying a world which he himself has not made and which would be present, essentially unchanged, if he were not there. Therefore, the Copenhagen interpretation offers no real understanding of the atomic phenomena.

It is easily seen that what this criticism demands is again the old materialistic ontology. But what can be the answer from the point of view of the Copenhagen interpretation?

We can say that physics is a part of science and as such aims at a description and understanding of nature. Any kind of understanding, scientific or not, depends on our language, on the communication of ideas. Every description of phenomena, of experiments and their results, rests upon language as the only means of communication. The words of this language represent the concepts of daily life, which in the scientific language of physics may be refined to the concepts of classical physics. These concepts are the only tools for an unambiguous communication about events, about the setting up of experiments and about their results. If therefore the atomic physicist is asked to give a description of what really happens in his experiments, the words "description" and "really" and "happens" can only refer to the concepts of daily life or of classical physics. As soon as the physicist gave up this basis he would lose the means of unambiguous communication and could not continue in his science. Therefore, any statement about what has "actually

happened” is a statement in terms of the classical concepts and—because of thermodynamics and of the uncertainty relations—by its very nature incomplete with respect to the details of the atomic events involved. The demand to “describe what happens” in the quantum-theoretical process between two successive observations is a contradiction *in adjecto*, since the word “describe” refers to the use of the classical concepts, while these concepts cannot be applied in the space between the observations; they can only be applied at the points of observation.

It should be noticed at this point that the Copenhagen interpretation of quantum theory is in no way positivistic. For, whereas positivism is based on the sensual perceptions of the observer as the elements of reality, the Copenhagen interpretation regards things and processes which are describable in terms of classical concepts, i.e., the actual, as the foundation of any physical interpretation.

At the same time we see that the statistical nature of the laws of microscopic physics cannot be avoided, since any knowledge of the “actual” is—because of the quantum-theoretical laws—by its very nature an incomplete knowledge.

The ontology of materialism rested upon the illusion that the kind of existence, the direct “actuality” of the world around us, can be extrapolated into the atomic range. This extrapolation is impossible, however.

A few remarks may be added concerning the formal structure of all the counterproposals hitherto made against the Copenhagen interpretation of quantum theory. All these proposals have found themselves compelled to sacrifice the essential symmetry properties of quantum theory (for instance, the symmetry between waves and particles or between position and velocity).

Therefore, we may well suppose that the Copenhagen interpretation cannot be avoided if these symmetry properties—like the Lorentz invariance in the theory of relativity—are held to be a genuine feature of nature; and every experiment yet performed supports this view.